SOFTWARE REQUIREMENTS SPECIFICATION

for

CAMPUSPULSE WEBSITE

Version 1.0

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1 Introduction

1.1 Purpose

In large academic institutions such as Thapar Institute of Engineering & Technology, maintaining and disseminating event information, club announcements, and workshop schedules through physical noticeboards or fragmented chat groups is highly error-prone and inefficient. Any event schedule, venue, or speaker can be changed, postponed, or added at any time. Freshmen and general students routinely face navigation confusion and miss out on important academic and extra-curricular opportunities. So, "CAMPUSPULSE WEBSITE" is the solution. "CAMPUSPULSE WEBSITE" is an official campus-wide web platform based on interactive event discovery, spatial map navigation, and 3-tier privacy calendar management authorized for TIET. The main concept of "CAMPUSPULSE WEBSITE" is to centralize all student societies, events, campus buildings, and peer schedules into an interconnected digital layer.

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, project managers, student coordinators, faculty evaluators, and software testers. Further the discussion will provide all the internal, external, functional, architectural, and also non-functional specifications about "CAMPUSPULSE WEBSITE". Developers should focus on the data schema and API specs, while project evaluators should inspect the scope, feature priorities, and user activity flows.

1.3 Project Scope

"CAMPUSPULSE WEBSITE" creates an integrated digital space for Students, Society Coordinators, and Campus Administrators for maintaining campus-wide social and academic interactions.

After registering on the platform using university credentials (roll number, email), a student can enter their personalized dashboard. The student can explore live upcoming events categorized by tags (Tech, Cultural, Sports, Hackathons, Free Food, Cash Prize), view exact geolocation pins on an interactive Leaflet campus map, and open sliding bottom drawers for walking directions. The student can RSVP to events and assign privacy tiers (Public, Friends Only, Private) to sync with their personal and friend group calendar.

Society Coordinators create official society profiles, post new event cards with date, time, venue pins, and incentive tags, and broadcast high-priority story reels. They also track real-time RSVP counts and attendee headcount.

Administrators oversee the integrity of the platform. Their main work is to verify society accounts, moderate event posts, manage campus landmark coordinates, and monitor system performance logs.

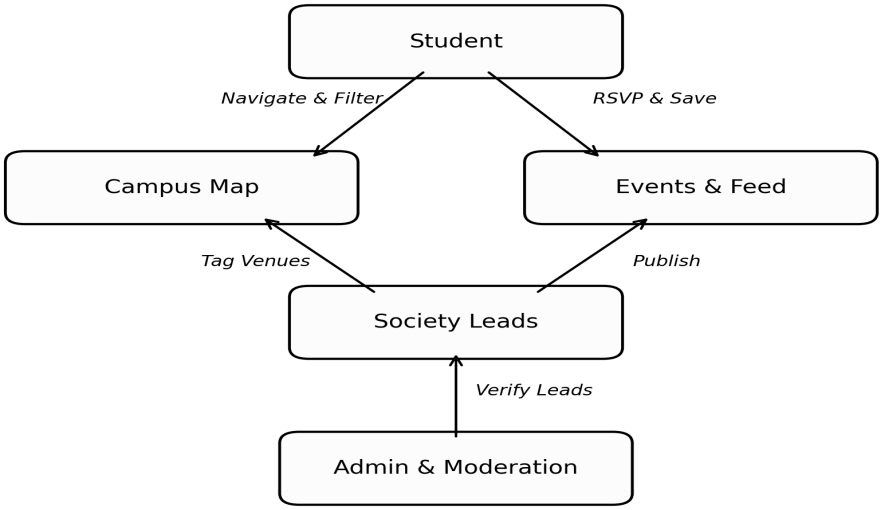


Figure 1.1: Entire work-flow

Figure 1.1 (Entire work-flow) is the overview of the project. Connection of all the entities are dependable on each other. This gives a clear and simple overview of the functional activities of the project.

Student cycle: In Figure 1.1, "Student" discovers "Events & Feed"; "Events" are linked to "Campus Map" landmarks; student RSVPs and synchronizes them to their "Social Calendar" with 3-tier privacy.

Society Lead cycle: "Society Leads" create and publish "Events"; tag specific geo-coordinates on "Campus Map"; broadcast announcement stories; and analyze live attendee metrics.

Administrator cycle: "Admin" audits submitted society events; moderates inappropriate content; maintains campus coordinate databases; and ensures system stability.

So, every entity is highly interactive and coupled in real-time with each other.

2 Overall Description

2.1 Product Perspective

"CAMPUSPULSE WEBSITE" is the replacement of manual paper posters, chaotic WhatsApp group blasts, and fragmented society notifications. All historical and upcoming event data, venue coordinates, and student schedules are stored centrally in a high-performance database. The main goal of this project is to minimize student friction, avoid schedule collisions, and maximize student participation in campus activities.

2.2 User Classes and Characteristics

"CAMPUSPULSE WEBSITE" has basically 3 primary types of users with distinct privileges:

• **Students**: Undergraduate and postgraduate campus students exploring events, syncing calendars, and navigating campus.
• **Society Leads / Coordinators**: Authorized club representatives creating events, broadcasts, and managing registrations.
• **Campus Administrators**: University authority maintaining map data, moderating posts, and approving society roles.

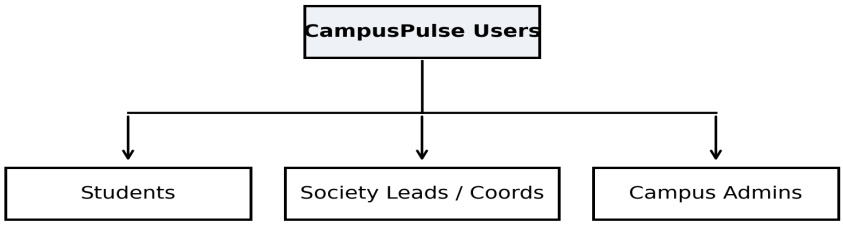


Figure 2.1: Type of users

2.3 Product Functions

"CAMPUSPULSE WEBSITE" stores and indexes all campus events, society broadcasts, venues, and student RSVPs. Before using the core features of the software, users must be registered and authenticated.

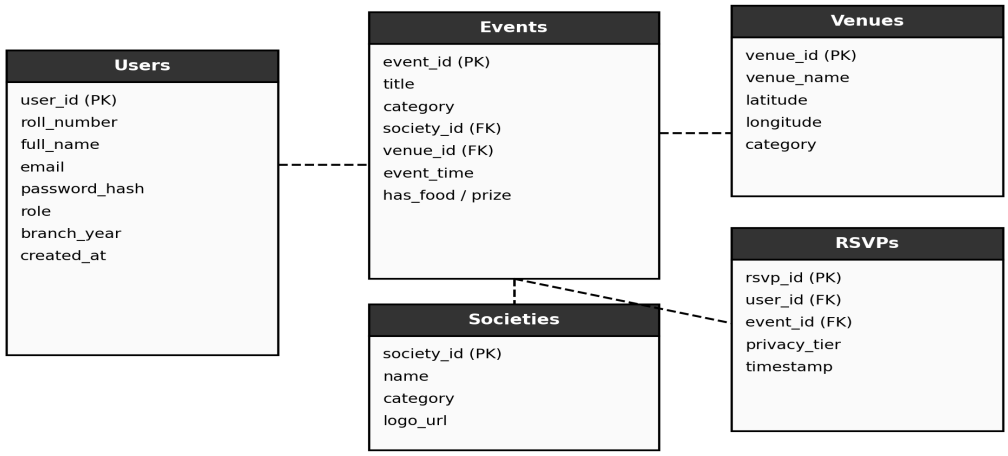


Figure 2.2: Data Flow and Entity Relationship Diagram

All users have standard attributes: user_id, roll_number, full_name, email, role, branch_year, password_hash, and created_at.

Events table contains: event_id, title, description, category (Tech, Cultural, Sports, Hackathon), society_id, venue_id, event_time, has_food, has_prize, and status.

Venues table contains: venue_id, venue_name, latitude, longitude, category (Academic, Hostel, Sports, Cafeteria), and description.

RSVP table is the core coordination bridge: rsvp_id, user_id, event_id, privacy_tier (Public, Friends Only, Private), and timestamp.

2.4 Operating Environment

The website will operate smoothly in any modern Operating Environment and Web Browser — Windows, macOS, Linux, Android, iOS, Google Chrome, Safari, Mozilla Firefox, and Microsoft Edge.

2.5 Design

Student activities follow 3 sequential steps:

• **Authentication & Profile**: Student enters roll number credentials and personal preferences.
• **Discovery & Navigation**: Student filters events by tags and selects map landmarks for directions.
• **RSVP & Privacy Sync**: Student saves events and assigns privacy tiers to coordinate with friends.



Figure 2.3: Student Activities

Every student profile displays their personal info, saved upcoming events, society memberships, and personalized friend feed. Notice and Story reels present urgent announcements and live registrations across Thapar.

Society Lead and Admin activities are structured as follows:

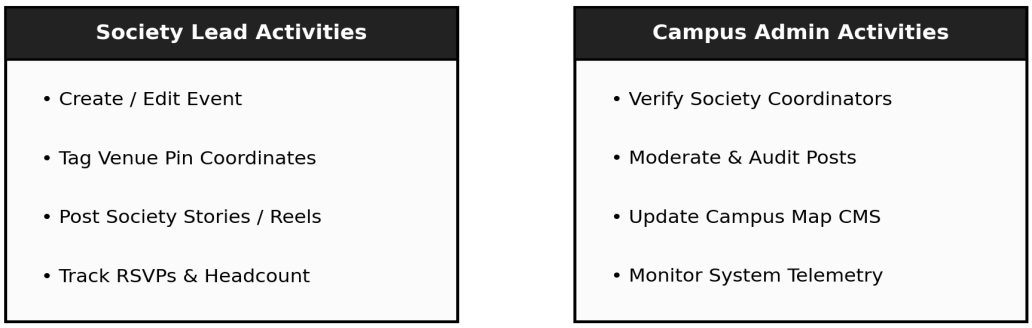


Figure 2.4: Society Coordinator and Campus Admin Activities

Society Coordinators: Authorized leads can create new events, specify exact campus coordinates, publish priority broadcasts/stories, and inspect live RSVP counts.

Campus Administrators: Administrators verify society leads, moderate all publicly posted events, manage landmark data on the map engine, and audit system security.

3 System Features

"CAMPUSPULSE WEBSITE" is a campus-wide event discovery and social calendar software. The core purpose of this product is to discover events, navigate campus spaces, and coordinate schedules effortlessly.

3.1 Description and Priority

The features with priority from highest to lowest:

1. **Interactive Event Feed**: Goal feature. Browse upcoming events, filter by categories, and search.
2. **Campus Geo-Spatial Map**: Interactive map with building markers and sliding bottom sheet navigation.
3. **3-Tier Privacy Calendar**: Personal schedule manager with Public, Friends Only, and Private tiers.
4. **Society Stories & Broadcasts**: Urgent announcement reel operated by verified club coordinators.
5. **Role-Based Authentication**: Secure JWT login and personal user profile management.
6. **Administrative Moderation**: Reviewing posts and managing campus landmark coordinates.

3.2 Functional Requirements & Tech Stack

The "CAMPUSPULSE WEBSITE" is built on modern high-performance web technologies:

• **Front-End**: React 19, Vite, Tailwind CSS, Framer Motion, Lucide Icons, Leaflet.js.
• **Back-End**: Python 3.11+, FastAPI framework, SQLAlchemy 2.0 (Async), Uvicorn.
• **Database**: SQLite (development) and PostgreSQL (production).
• **UI Theme**: Dark Y2K Neon aesthetic (#07070b dark canvas with cyan & purple accents).

4 Other Nonfunctional Requirements

4.1 Performance Requirements

"CAMPUSPULSE WEBSITE" is engineered for high-concurrency peak usage during fest registrations and hackathons. First Contentful Paint (FCP) shall be < 1.5 seconds on 4G networks. Core API responses shall be served under 100ms.

4.2 Security Requirements

No unregistered or unauthorized user can create events or modify campus landmarks. Passwords must be hashed using bcrypt before database storage. Communication is enforced over HTTPS/TLS.

4.3 Software Quality Attributes

During all development phases, continuous automated unit tests, API integration tests, and UI component testing are maintained. The system maintains 99.5% uptime availability during active academic terms.

4.4 Business Rules

• **Verification**: Only verified society coordinator accounts can publish official events.
• **Privacy Default**: New RSVPs default to Friends Only visibility.
• **Archival**: Expired events automatically transition to the archive state 24 hours after completion.

5 Other Requirements

"CAMPUSPULSE WEBSITE" needs ongoing maintenance as it is a multi-semester evolving software system. It features a modular code architecture to support continuous refactoring as new university requirements and campus buildings emerge.

Future Enhancements:

• **Turn-by-turn Campus Pathfinding**: Integration of Dijkstra/A* pathfinding algorithms for walking paths between hostels, academic blocks, and food courts.

• **Push Notifications**: Real-time service worker notifications for registered event reminders and schedule updates.

• **Multi-Campus Expansion**: Modular schema extension allowing multi-tenant deployment across other universities.